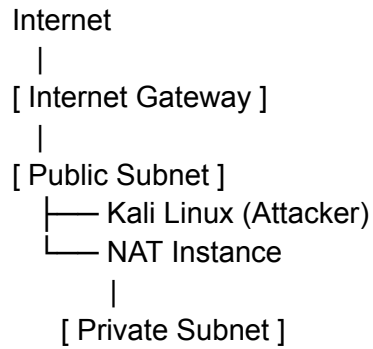
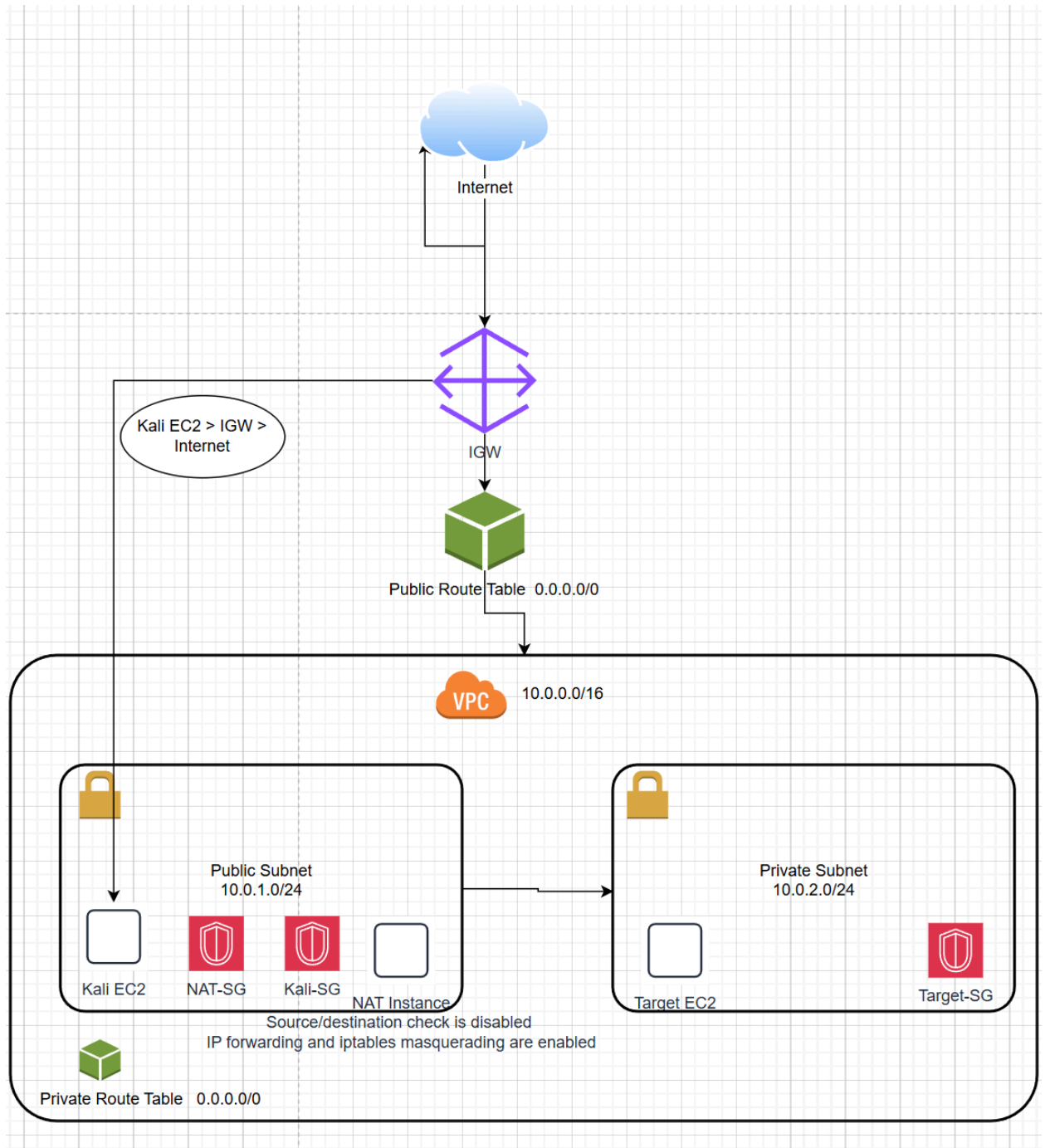




- Design the network
- Deploy Kali linux
- Create a NAT Instance (Free Tier)

This project demonstrates how to design and deploy a secure AWS VPC architecture using public and private subnets, a NAT instance, and a Kali Linux EC2 for controlled testing. It showcases cloud security fundamentals, network segmentation, Linux hardening, and secure outbound routing using free-tier resources.



Network Foundation

1. Create the VPC
 - AWS Console > VPC > Create VPC
 - Name: Cloud-Security-VPC
 - IPv4 CIDR: 10.0.0.0/16
 - Tenancy: Default

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)

Create only the VPC resource or the VPC and other networking resources.

☒ VPC only

☐ VPC and more

Name tag - *optional*

Creates a tag with a key of 'Name' and a value that you specify.

Cloud-Security-VPC

IPv4 CIDR block [Info](#)

☒ IPv4 CIDR manual input

☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

10.0.0.0/16

CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)

☒ No IPv6 CIDR block

☐ IPAM-allocated IPv6 CIDR block

☐ Amazon-provided IPv6 CIDR block

☐ IPv6 CIDR owned by me

Tenancy [Info](#)

Default

2. Create Subnets

Public Subnet

- Name: Public-Subnet
- CIDR: 10.0.1.0/24
- Enable auto-assign public IP

Create subnet [Info](#)

VPC

VPC ID

Create subnets in this VPC.

vpc-0889fd29c004efe3b (Cloud-Security-VPC) ▼

Associated VPC CIDRs

IPv4 CIDRs

10.0.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Public-Subent

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (Ohio) / use2-az1 (us-east-2a) ▼

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16 ▼

IPv4 subnet CIDR block

10.0.1.0/24

256 IPs

< > ^ v

Edit subnet settings [Info](#)

Subnet

Subnet ID

 subnet-02df60a62f408a485

Name

 Public-Subent

Auto-assign IP settings [Info](#)

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign public IPv4 address [Info](#)

☐ Enable auto-assign customer-owned IPv4 address [Info](#)
Option disabled because no customer owned pools found.

Private Subnet

- Name: Private-Subnet
- CIDR: 10.0.2.0/24
- Auto-assign public IP: Disabled

Create subnet [Info](#)

VPC

VPC ID

Create subnets in this VPC.

vpc-0889fd29c004efe3b (Cloud-Security-VPC) ▼

Associated VPC CIDRs

IPv4 CIDRs

10.0.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Private-Subnet

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (Ohio) / use2-az1 (us-east-2a) ▼

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16 ▼

IPv4 subnet CIDR block

10.0.2.0/24

256 IPs

< > ^ v

Tags - optional

3. Internet Gateway

- Create IG: Cloud-IG
- Attach to Cloud-Security-VPC

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag

Creates a tag with a key of 'Name' and a value that you specify.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key



Value - optional

[Remove](#)[Add new tag](#)

You can add 49 more tags.

[Cancel](#)[Create internet gateway](#)

Attach to VPC (igw-04ad395a3b08377f5) [Info](#)

VPC

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs

Attach the internet gateway to this VPC.



Use: "vpc-0889fd29c004efe3b"

vpc-0889fd29c004efe3b - Cloud-Security-VPC

4. Route Tables

Public Route Table

Routes: 0.0.0.0/0 > IG

Associate: Public-Subnet

VPC
>
Route tables
>
rtb-082521d9a5e29dd89
>
Edit routes

Destination
10.0.0.0/16

Target
local

Status
Active

Propagated
No

Route Origin
CreateRouteTable

0.0.0.0/0

Internet Gateway

-

No

CreateRoute

Remove

Add route

Cancel
Preview
Save changes

You have successfully updated subnet associations for rtb-082521d9a5e29dd89 / Public Route Table.

rtb-082521d9a5e29dd89 / Public Route Table
Actions

Details
Info

Route table ID
rtb-082521d9a5e29dd89

VPC
vpc-0889fd29c004efe3b | Cloud-Security-VPC

Main
No

Owner ID
277848663122

Explicit subnet associations
subnet-02df60a62f408a485 / Public-Subent

Edge associations
-

Security Controls

- 5. Security Groups
 - Kali-SG
 - Inbound: SSH (22) > /32
 - Outbound: Allow all

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Kali-SG

Name cannot be edited after creation.

Description [Info](#)

Kali Linux

VPC [Info](#)

vpc-0889fd29c004efe3b (Cloud-Security-VPC)

Inbound rules [Info](#)

Type [Info](#)

Protocol

Port range [Info](#)

Source [Info](#)

Description - optional [Info](#)

SSH

TCP

22

My IP

Q

0.0.0.0/0

Delete

Add rule

Outbound rules [Info](#)

Type [Info](#)

Protocol

Port range [Info](#)

Destination [Info](#)

Description - optional [Info](#)

All traffic

All

All

Cu...

Q

0.0.0.0/0

Delete

Add rule

NAT-SG

Inbound:

- SSH > /32
- ALL traffic > 10.0.2.0/24

Outbound:

- Allow all

Info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

NAT-SG

Name cannot be edited after creation.

Description Info

NAT

VPC Info

vpc-0889fd29c004efe3b (Cloud-Security-VPC)

Inbound rules [Info](#)

Type Info

Protocol

Port range [Info](#)

Source [Info](#)

Description - optional [Info](#)

SSH

TCP

22

My IP ▼

Delete

All traffic

All

All

Cu... ▼

🔍 10.0.2.0/24

Q

Delete

Add rule

Outbound rules [Info](#)

Type Info

Protocol

Port range [Info](#)

Destination Info

Description - optional [Info](#)

All traffic

All

All

An... ▼

0.0.0.0/0

□

Delete

Add rule

0.0.0.0/0

Target-SG

Inbound: All traffic > 10.0.1.0/24

Outbound: Allow all

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Name cannot be edited after creation.

Description [Info](#)

VPC [Info](#)

Inbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info
All traffic ▼	All	All	Cu... Q 10.0.1.0/24 X 10.0.1.0/24 X	
Delete				
Add rule				

Outbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Destination Info	Description - optional Info
All traffic ▼	All	All	Cu... Q 0.0.0.0/0 X	
Delete				
Add rule				

Deploy Kali Linux (Red Team)

- Launch Kali Linux EC2
 - AMI: Official Kali Linux
 - Instance Type: t2.micro
 - Subnet: Public-Subnet
 - Auto-assign public IP: Enabled
 - Security Group: Kali-SG
 - Storage: 20-30 GB

Amazon Machine Images (AMIs) (1/6) Info



Recycle Bin

EC2 Image Builder

Actions

Launch instance from AMI

Public images

Search

Kali Linux



Clear filters

< 1 > ⚙

	Name	AMI name	AMI ID	Source	Owr
☐		kali-last-snapshot-arm64-2025....	ami-0350e4410f8426251	aws-marketplace/kali-last-snapshot-ar...	679!
☒		kali-last-snapshot-amd64-2025...	ami-0723a2f2ccd67a503	aws-marketplace/kali-last-snapshot-am...	679!
☐		kali-last-snapshot-amd64-2025...	ami-07e20b1379c448040	aws-marketplace/kali-last-snapshot-am...	679!
☐		kali-last-snapshot-arm64-2025....	ami-0ad66086ac209d4af	aws-marketplace/kali-last-snapshot-ar...	679!
☐		Kali Linux -AWS-Nuvemnest-pro...	ami-0aa78ade27eae9e0d	aws-marketplace/Kali Linux -AWS-Nuve...	679!
☐		Kali Linux On AWS-239c5ea9-c...	ami-0b94eb86457a5508a	aws-marketplace/Kali Linux On AWS-23...	679!

AMI ID: ami-0723a2f2ccd67a503



Details

Storage

AMI ancestry - new

Tags

AMI ID ami-0723a2f2ccd67a503	Image type machine	Platform details Linux/UNIX	Root device type EBS
AMI name kali-last-snapshot-amd64-2025.4.0-804fcc46-63fc-4eb6-85a1-50e66d6c7215	Owner account ID 67959333241	Architecture x86_64	Usage operation RunInstances
Root device name /dev/xvda	Status Available	Source aws-marketplace/kali-last-snapshot-amd64-2025.4.0-804fcc46-63fc-4eb6-85a1-50e66d6c7215	Virtualization type hvm
Boot mode -	State reason -	Creation date 2025-12-19T17:18:03.000Z	Kernel ID -
Description Kali Linux kali-last-snapshot (2025.4.0)	Product codes marketplace:7lgvy7mt78lgoi4lant0znp5h	RAM disk ID -	Deprecation time Sun Dec 19 2027 11:18:03 GMT-0600 (Central Standard Time)

[AMI from catalog](#)[Quick Start](#)

Name

kali-last-snapshot-amd64-2025.4.0-804fcc46-63fc-4eb6-85a1-50e66d6c7215

Verified provider

Free tier eligible

[Browse more AMIs](#)

Including AMIs from
AWS, Marketplace and
the Community

Description

Kali Linux kali-last-snapshot (2025.4.0)

Image ID

ami-0723a2f2ccd67a503

Username ⓘ

root (Check with the AMI provider.)

Catalog	Published	Architecture	Virtualization	Root device type	ENA Enabled
AWS Marketplace AMIs	2025-12-19T17:18:03.00Z	x86_64	hvm	ebs	Yes

If you have an existing license entitlement to use this software, then you can launch this software without creating a new subscription. If you do not have an existing entitlement, then by launching this software, you will be subscribed to this software and agree that your use of this software is subject to the pricing terms and the seller's [End User License Agreement](#)

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro

Free tier eligible ▼

Family: t3 2 vCPU 1 GiB Memory Current generation: true

☒ All generations[Compare instance types](#)

The AMI vendor recommends using a t2.medium instance (or larger) for the best experience with this product.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

cloud-security-kali-key ▼

[Create new key pair](#)▼ Network settings [Info](#)VPC - *required* | [Info](#)

vpc-0889fd29c004efe3b (Cloud-Security-VPC)
10.0.0.0/16 ▼

Subnet | [Info](#)

subnet-02df60a62f408a485 Public-Subent
VPC: vpc-0889fd29c004efe3b Owner: 277848663122
Availability Zone: us-east-2a (use2-az1) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.0.1.0/24 ▼

[Create new subnet](#)

Auto-assign public IP | [Info](#)

Enable ▼

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group☒ Select existing security group**Common security groups** | [Info](#)

Select security groups ▼

Kali-SG sg-0ace02431f042c4ad ✕
VPC: vpc-0889fd29c004efe3b

[↻ Compare security group rules](#)

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration**▼ Configure storage** | [Info](#)[Advanced](#)

1x 25 GiB gp2 ▼ Root volume, Not encrypted

SSH: `ssh -i kali.pem kali@<Public-IP>`

```
C:\Users\ndr\Downloads>ssh -i C:\Users\ndr\Downloads\cloud-security-kali-key.pem kali@18.227.111.248
The authenticity of host '18.227.111.248 (18.227.111.248)' can't be established.
ED25519 key fingerprint is SHA256:ABye8cCWUwBR+V0sNdmSsiaNizkODvVXDqSOCGPre2Q.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '18.227.111.248' (ED25519) to the list of known hosts.
Linux kali 6.16.8+kali-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.16.8-1kali1 (2025-09-24) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
(Message from Kali developers)

This is a minimal installation of Kali Linux, you likely
want to install supplementary tools. Learn how:
= https://www.kali.org/docs/troubleshooting/common-minimum-setup/

This is a cloud installation of Kali Linux. Learn more about
the specificities of the various cloud images:
= https://www.kali.org/docs/troubleshooting/common-cloud-setup/

(Run: "touch ~/.hushlogin" to hide this message)
(kali@kali)~$
```

Verify:

Ping google.com

```

(kali㉿kali)-[~]
└─$ ping google.com
PING google.com (142.250.191.142) 56(84) bytes of data:
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=1 ttl=117 time=8.69 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=2 ttl=117 time=8.79 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=3 ttl=117 time=8.70 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=4 ttl=117 time=8.71 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=5 ttl=117 time=8.70 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=6 ttl=117 time=8.69 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=7 ttl=117 time=8.72 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=8 ttl=117 time=8.78 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=9 ttl=117 time=8.72 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=10 ttl=117 time=8.80 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=11 ttl=117 time=8.72 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=12 ttl=117 time=8.70 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=13 ttl=117 time=8.71 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=14 ttl=117 time=8.72 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=15 ttl=117 time=8.71 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=16 ttl=117 time=8.72 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=17 ttl=117 time=8.73 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=18 ttl=117 time=8.69 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=19 ttl=117 time=8.69 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=20 ttl=117 time=8.72 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=21 ttl=117 time=8.71 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=22 ttl=117 time=8.70 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=23 ttl=117 time=8.73 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=24 ttl=117 time=8.70 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=25 ttl=117 time=8.71 ms
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=26 ttl=117 time=8.70 ms

^C
i--- google.com ping statistics ---
61 packets transmitted, 61 received, 0% packet loss, time 60113ms
rtt min/avg/max/mdev = 8.685/8.714/8.797/0.022 ms

```

NAT Instance (FREE TIER MAGIC)

7. Launch NAT Instance
 - AMI: Amazon Linux 2
 - Instance Type: t2.micro
 - Subnet: Public-subnet
 - Auto-assign Public IP: Enabled
 - Security Groups: NAT-SG

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

nat-instance-public

[Add additional tags](#)

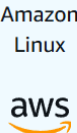
▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

 Search our full catalog including 1000s of application and OS images

Recents

Quick Start



Amazon Linux



macOS



Ubuntu



Windows



Red Hat



SUSE Linu



[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-06f1fc9ae5ae7f31e (64-bit (x86), uefi-preferred) / ami-058e74ab207ed2b33 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.10.20260105.0 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username	
64-bit ... 	uefi-preferred	ami-06f1fc9ae5ae7f31e	2026-01-02	ec2-user	

Verified provider

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro

Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand RHEL base pricing: 0.0392 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour
On-Demand Linux base pricing: 0.0104 USD per Hour

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

cloud-security-kali-key

[Create new key pair](#)

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-0889fd29c004efe3b (Cloud-Security-VPC)
10.0.0.0/16



Subnet [Info](#)

subnet-02df60a62f408a485 Public-Subnet
VPC: vpc-0889fd29c004efe3b Owner: 277848663122
Availability Zone: us-east-2a (use2-az1) Zone type: Availability Zone
IP addresses available: 250 CIDR: 10.0.1.0/24



[Create new subnet](#)

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

Common security groups [Info](#)

Select security groups



[Compare security group rules](#)

NAT-SG sg-01d741fc8ae3c57a3 X
VPC: vpc-0889fd29c004efe3b

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

▼ Configure storage [Info](#)

Advanced

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

EC2 > Instance > Networking > Disable

Enable Forwarding

Persist:

```
C:\Users\MGFel\Downloads>cloud-security-kali-key.pem

C:\Users\MGFel\Downloads>ssh -i cloud-security-kali-key.pem ec2-user@3.143.7.149

#
~\##### Amazon Linux 2023
~\#####
~\###|
~\#| https://aws.amazon.com/linux/amazon-linux-2023
~\V~'-'>
~\
~\
~\
~\m/'
```

```
net.ipv4.ip_forward=1
```

```
ec2-user@ip-10-0-1-172:~
```

```
GNU nano 8.3
```

```
/etc/sysctl.conf
```

```
net.ipv4.ip_forward = 1
```

Configure iptables

```
sudo iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
```

```
sudo yum install iptables-services -y
```

```
sudo service iptables save
```

```

sudo: iptables: command not found
[ec2-user@ip-10-0-1-172 ~]$ sudo yum install iptables-services -y
Amazon Linux 2023 Kernel Livepatch repository                248 kB/s | 30 kB    00:00
Last metadata expiration check: 0:00:01 ago on Thu Jan  8 02:47:22 2026.
Dependencies resolved.
=====
Package                        Architecture      Version           Repository         Size
=====
Installing:
iptables-services              noarch            1.8.8-3.amzn2023.0.2  amazonlinux        18 k
Installing dependencies:
iptables-libs                  x86_64            1.8.8-3.amzn2023.0.2  amazonlinux        401 k
iptables-nft                   x86_64            1.8.8-3.amzn2023.0.2  amazonlinux        183 k
iptables-utils                 x86_64            1.8.8-3.amzn2023.0.2  amazonlinux        43 k
libnetfilter_conntrack         x86_64            1.0.8-2.amzn2023.0.2  amazonlinux        58 k
libnftnl                       x86_64            1.0.1-19.amzn2023.0.2  amazonlinux        30 k
libnftnl                       x86_64            1.2.2-2.amzn2023.0.2  amazonlinux        84 k
=====
Transaction Summary
=====
Install 7 Packages

Total download size: 816 k
Installed size: 2.9 M
Downloading Packages:
(1/7): iptables-services-1.8.8-3.amzn2023.0.2.noarch.rpm    495 kB/s | 18 kB    00:00
(2/7): iptables-libs-1.8.8-3.amzn2023.0.2.x86_64.rpm      8.4 MB/s | 401 kB   00:00
(3/7): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm       3.4 MB/s | 183 kB   00:00
(4/7): iptables-utils-1.8.8-3.amzn2023.0.2.x86_64.rpm     1.5 MB/s | 43 kB    00:00
(5/7): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm 1.9 MB/s | 58 kB    00:00
(6/7): libnftnl-1.0.1-19.amzn2023.0.2.x86_64.rpm          1.0 MB/s | 30 kB    00:00
(7/7): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm           2.7 MB/s | 84 kB    00:00
-----
Total                                                    5.9 MB/s | 816 kB   00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing                : 1/1
  Installing               : libnftnl-1.0.1-19.amzn2023.0.2.x86_64 1/7
  Installing               : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64 2/7
  Installing               : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64 3/7
  Installing               : libnftnl-1.2.2-2.amzn2023.0.2.x86_64 4/7
  Installing               : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64 5/7
  Running scriptlet: iptables-nft-1.8.8-3.amzn2023.0.2.x86_64 5/7
  Installing               : iptables-utils-1.8.8-3.amzn2023.0.2.x86_64 6/7
  Installing               : iptables-services-1.8.8-3.amzn2023.0.2.noarch 7/7
  Running scriptlet: iptables-services-1.8.8-3.amzn2023.0.2.noarch 7/7
  Verifying               : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64 1/7
  Verifying               : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64 2/7
  Verifying               : iptables-services-1.8.8-3.amzn2023.0.2.noarch 3/7
  Verifying               : iptables-utils-1.8.8-3.amzn2023.0.2.x86_64 4/7
  Verifying               : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64 5/7
  Verifying               : libnftnl-1.0.1-19.amzn2023.0.2.x86_64 6/7
  Verifying               : libnftnl-1.2.2-2.amzn2023.0.2.x86_64 7/7
Installed:
iptables-libs-1.8.8-3.amzn2023.0.2.x86_64      iptables-nft-1.8.8-3.amzn2023.0.2.x86_64
iptables-services-1.8.8-3.amzn2023.0.2.noarch  iptables-utils-1.8.8-3.amzn2023.0.2.x86_64
libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64  libnftnl-1.0.1-19.amzn2023.0.2.x86_64
libnftnl-1.2.2-2.amzn2023.0.2.x86_64
Complete!
[ec2-user@ip-10-0-1-172 ~]$ iptables --version
iptables v1.8.8 (nf_tables)
[ec2-user@ip-10-0-1-172 ~]$ sudo systemctl start iptables
[ec2-user@ip-10-0-1-172 ~]$ sudo systemctl enable iptables
Created symlink /etc/systemd/system/multi-user.target.wants/iptables.service → /usr/lib/systemd/system/iptables.service.
[ec2-user@ip-10-0-1-172 ~]$ sudo iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
[ec2-user@ip-10-0-1-172 ~]$ sudo iptables -t nat -L -n -v
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target     prot opt in     out     source    destination

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target     prot opt in     out     source    destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target     prot opt in     out     source    destination

Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target     prot opt in     out     source    destination
    0      0 MASQUERADE all  --  *      eth0    0.0.0.0/0  0.0.0.0/0

```

```

[ec2-user@ip-10-0-1-172 ~]$ sudo service iptables save
iptables: Saving firewall rules to /etc/sysconfig/iptables: [ OK ]
[ec2-user@ip-10-0-1-172 ~]$ cat /proc/sys/net/ipv4/ip_forward
1
[ec2-user@ip-10-0-1-172 ~]$ sudo iptables -t nat -L
Chain PREROUTING (policy ACCEPT)
target     prot opt source                destination

Chain INPUT (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination

Chain POSTROUTING (policy ACCEPT)
target     prot opt source                destination
MASQUERADE all  --  anywhere              anywhere

```

8. Update Private Routing Table

0.0.0.0/0 > NAT Instance ID (Private subnet now has outbound internet without exposure)

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Network Interface	Active	No	CreateRoute

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

☐	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	Public-Subnet	subnet-02df60a62f408a485	10.0.1.0/24	-	rtb-082521d9a5e29dd89 / Public R
☒	Private-Subnet	subnet-05edc509e37269107	10.0.2.0/24	-	Main (rtb-0be16b26f8b06f1ad)

Selected subnets

[subnet-05edc509e37269107 / Private-Subnet](#) X

Cancel

Save associations

```
(kali㉿kali)-[~]
$ ping google.com
PING google.com (142.250.190.14) 56(84) bytes of data.
64 bytes from ord37s32-in-f14.1e100.net (142.250.190.14): icmp_seq=1 ttl=117 time=8.43 ms
64 bytes from ord37s32-in-f14.1e100.net (142.250.190.14): icmp_seq=2 ttl=117 time=8.43 ms
64 bytes from ord37s32-in-f14.1e100.net (142.250.190.14): icmp_seq=3 ttl=117 time=8.52 ms
64 bytes from ord37s32-in-f14.1e100.net (142.250.190.14): icmp_seq=4 ttl=117 time=8.48 ms
64 bytes from ord37s32-in-f14.1e100.net (142.250.190.14): icmp_seq=5 ttl=117 time=8.45 ms
64 bytes from ord37s32-in-f14.1e100.net (142.250.190.14): icmp_seq=6 ttl=117 time=8.44 ms
64 bytes from ord37s32-in-f14.1e100.net (142.250.190.14): icmp_seq=7 ttl=117 time=8.45 ms
64 bytes from ord37s32-in-f14.1e100.net (142.250.190.14): icmp_seq=8 ttl=117 time=8.45 ms
^C
--- google.com ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7011ms
rtt min/avg/max/mdev = 8.427/8.455/8.516/0.026 ms
```

Threat Model

Threats Considered

- Public subnet exposure
- SSH brute-force attempts
- NAT instance compromise
- Misconfigured route tables
- Lateral movement from public > private subnet
- Privilege escalation on EC2 instances
- Data exfiltration from private subnet

Mitigations

- SSH restricted top /32

- No public IPs in private subnet
- NAT instance hardened (IP forwarding, iptables)
- Strict Security Groups

Cost-Control Strategy

- All EC2 instances free-tier eligible (t2.micro/t3.micro)
- NAT instance used instead of NAT Gateway
- No load balancers, RDS, or high-cost services
- Instances stopped when not in use
- No attack traffic to avoid egress charges

This project demonstrates my ability to design secure cloud architectures, implement Linux-based NAT routing, enforce least-privilege network segmentation, and validate connectivity in AWS. It reflects my interest in cloud security engineering practices.

[Cloud-Security-Lab](#)